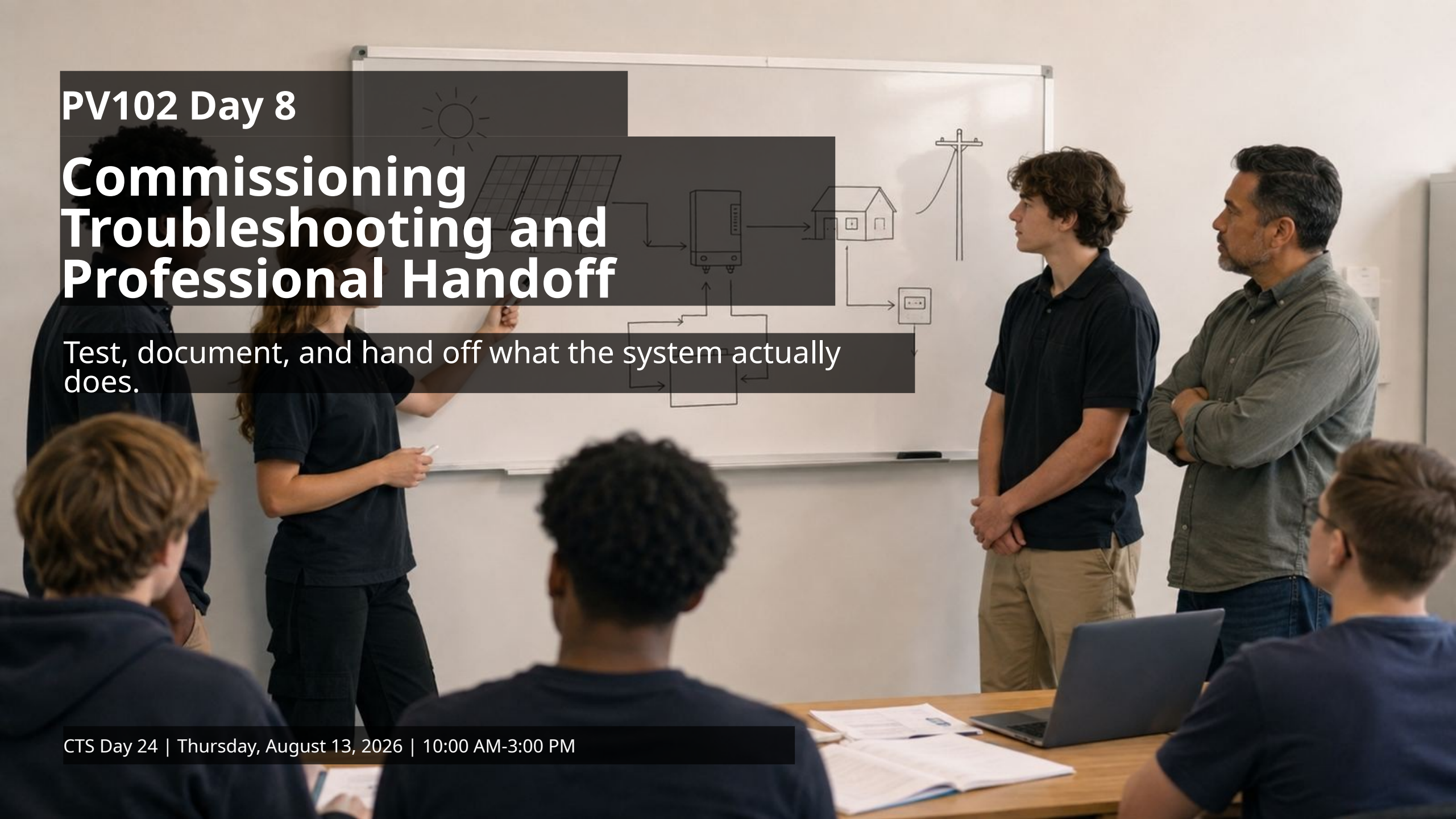


PV102 Day 8

Commissioning Troubleshooting and Professional Handoff

Test, document, and hand off what the system actually does.

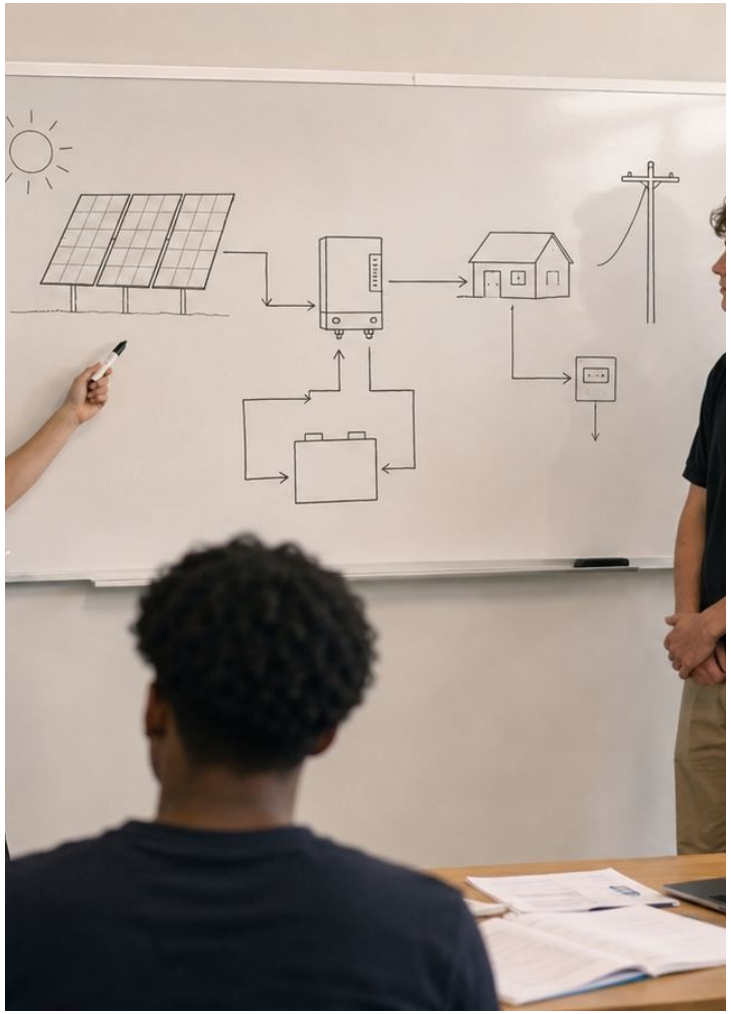
CTS Day 24 | Thursday, August 13, 2026 | 10:00 AM-3:00 PM

Today’s job

Professional solar work starts before parts move.

• Students close PV102 by verifying a system from evidence: pre-energization inspection, polarity and continuity checks, voltage checks, functional checks, fault response, documentation, and client handoff.

- Carry forward the Day 1 habit: match architecture to real need.
- Today adds one design layer and one clean handoff.



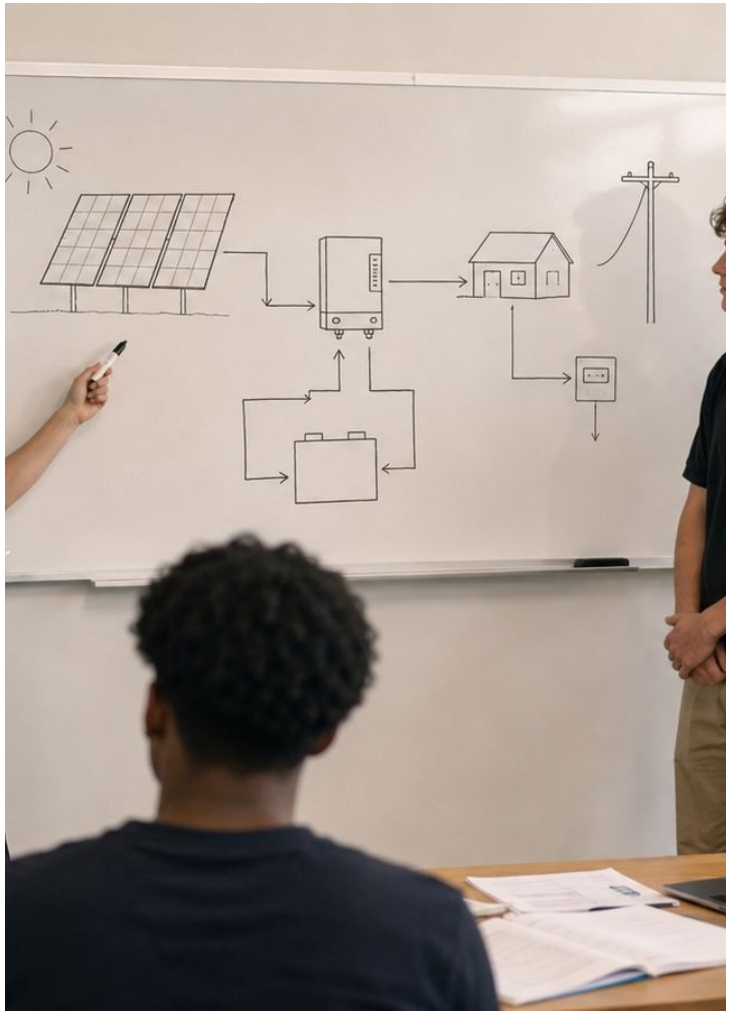
Do not guess when an unknown can be recorded.

Test, document, and hand off what the system actually does.

Learning targets

By 3:00 PM students should be able to:

- Use a commissioning sequence to verify before energizing.
- Interpret basic test results without guessing beyond evidence.
- Troubleshoot from symptom to boundary to measurement.
- Prepare a concise handoff that names what passed, what failed, what changed, and what remains unknown.



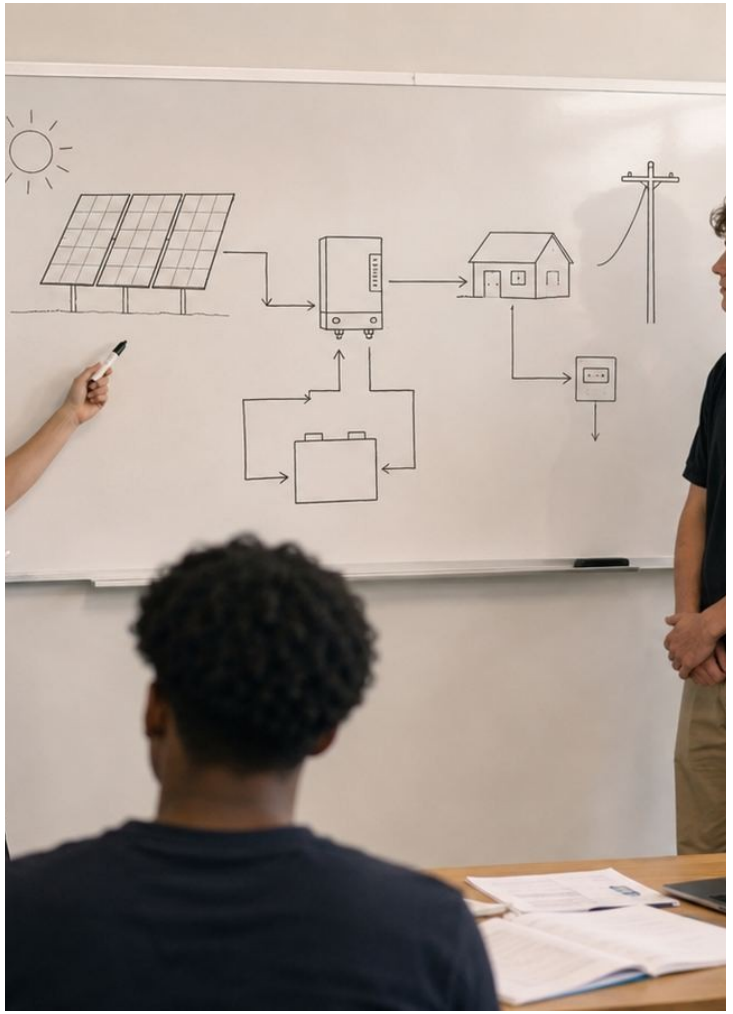
Evidence beats confidence. A tragic lesson for civilization, apparently.

Test, document, and hand off what the system actually does.

Morning readiness check

Quiz first, then review as evidence.

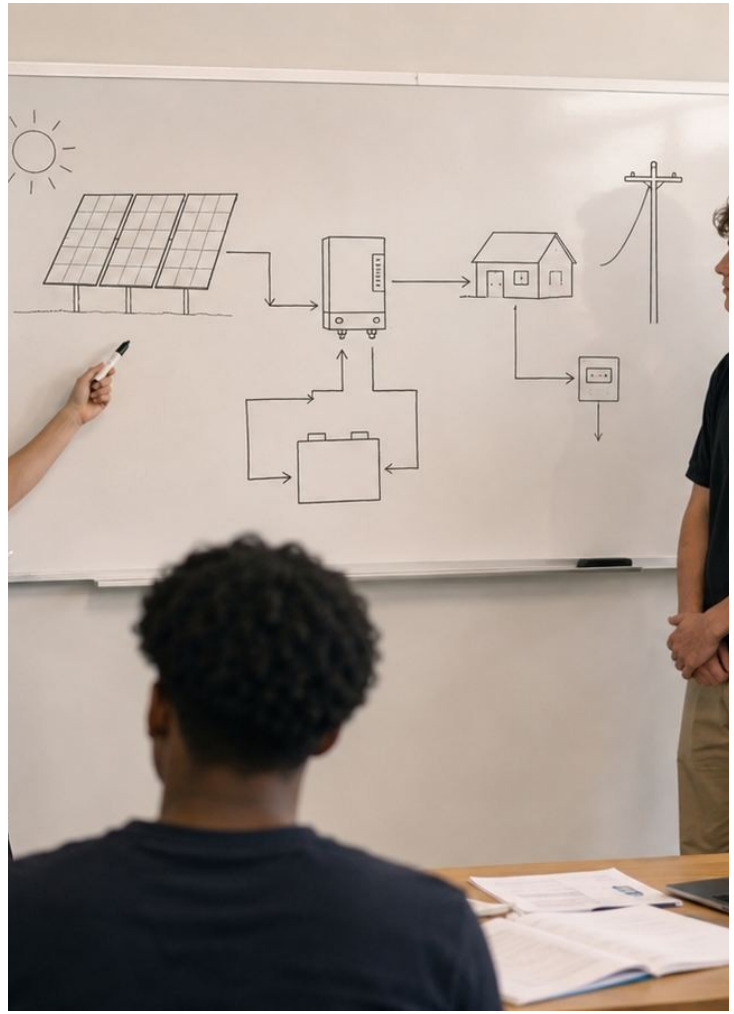
- Work independently for 12-15 minutes.
- Use wrong answers to locate teaching gaps.
- Ask: what fact would prove this answer?
- Keep review short and specific.



The quiz measures instruction as much as students.

Core rule

- Test, document, and hand off what the system actually does.
- Say it once at the start.
- Bring every debate back to it.
- Use it to reject weak handoffs.



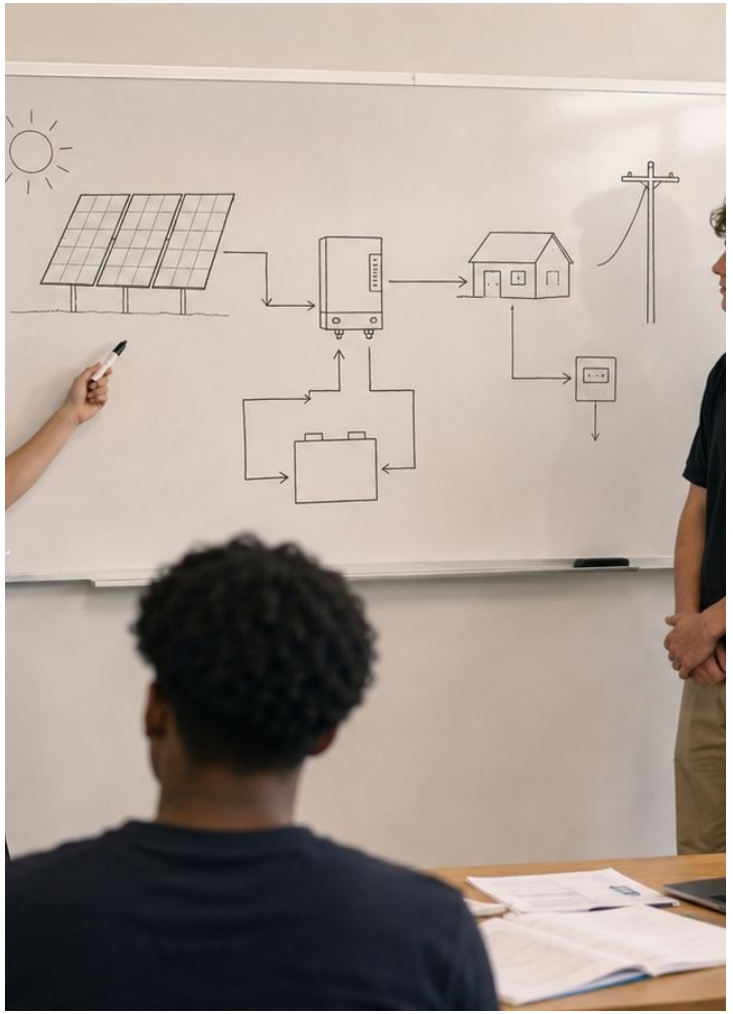
One rule, repeated until reality behaves.

Test, document, and hand off what the system actually does.

What students already know

PV101 and RCS are not separate islands.

- PV modules produce DC power.
- Series and parallel change voltage/current behavior.
- Continuity requires de-energized verification.
- Troubleshooting starts from evidence.
- Handoffs protect the next person.

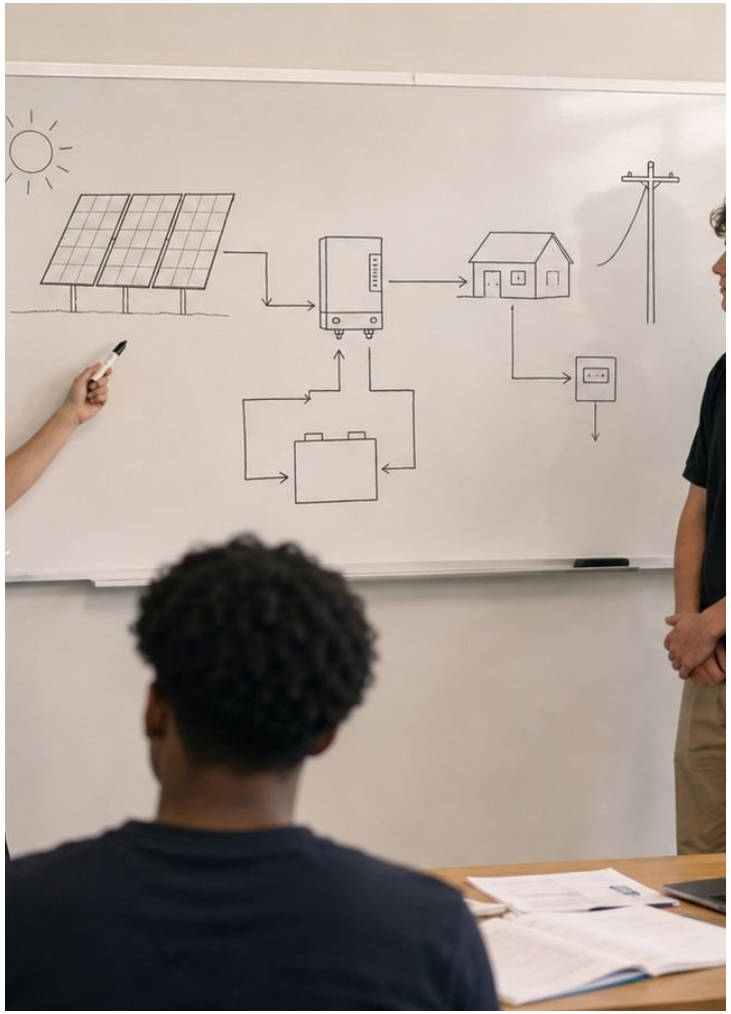


Test, document, and hand off what the system actually does.

New judgment today

The skill moves from “can I build it?” to “should this be the design?”

- Define the job.
- Gather the evidence.
- Name the constraint.
- Protect the boundary.
- Record unknowns.



Test, document, and hand off what the system actually does.

Energy path and decision boundary



Every arrow is a boundary. Boundaries need ratings, protection, isolation, and documentation.

Test, document, and hand off what the system actually does.

Decision checks

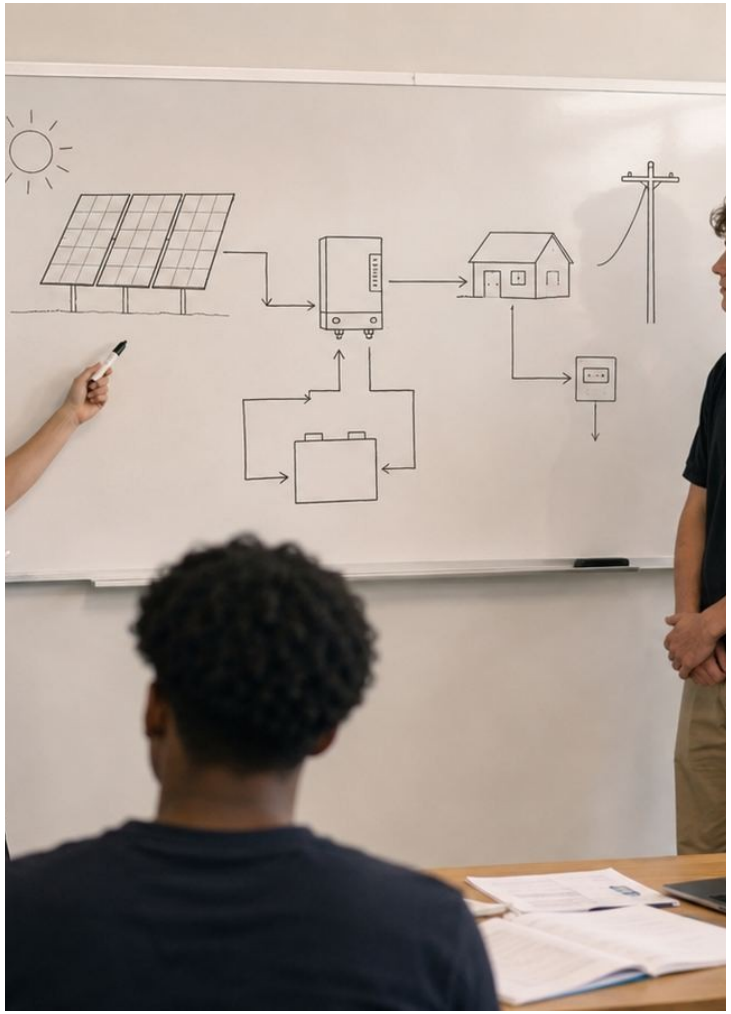
Decision	Evidence	Reject if
Keep a design choice	Measured or observed site/load data	It only sounds good
Reject a shortcut	Safety or equipment boundary	It hides a hazard
Carry unknown forward	Named verification step	It becomes fake certainty
Approve handoff	Next person can act safely	It depends on mind reading

Test, document, and hand off what the system actually does.

Guided example

Model one complete thinking path.

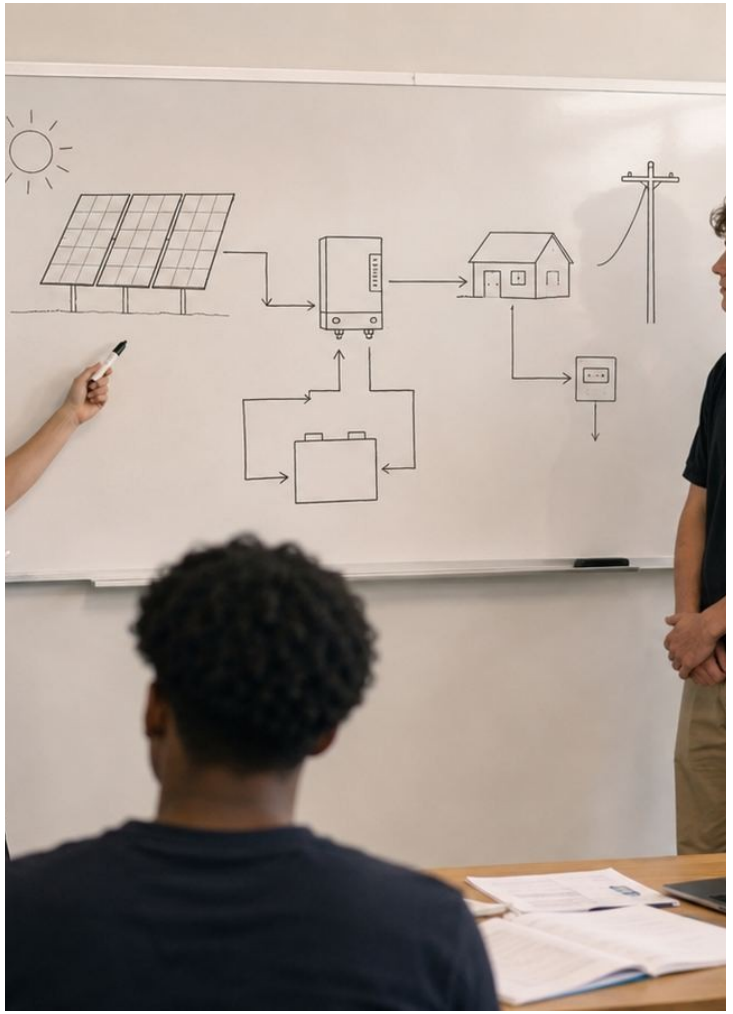
- Start with the client/job statement.
- Add one measured or observed fact.
- Make one design decision.
- Name one safety boundary.
- Name one unknown and how to verify it.



Team roles

Three teams of two, roles rotate after the first pass.

- Designer: proposes the first path.
- Verifier: challenges evidence and limits.
- Recorder/handoff lead: writes what the next person needs.
- In a pair, one person may hold two roles, then swap.

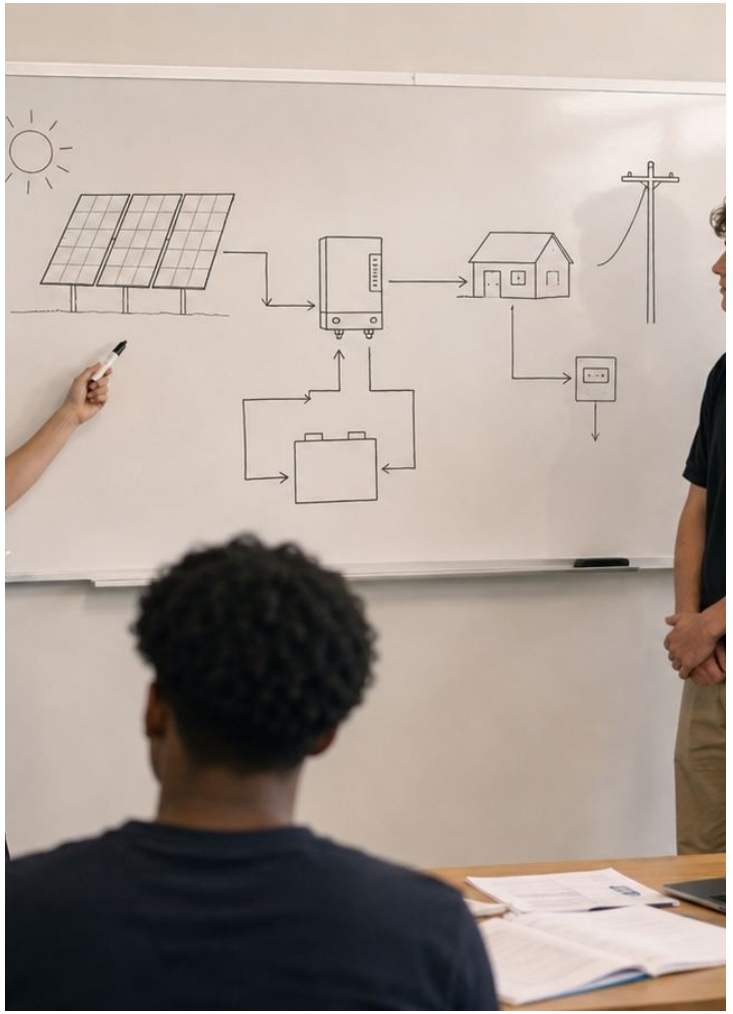


Test, document, and hand off what the system actually does.

Lab mission

Commissioning and Fault Handoff Challenge

- Teams inspect a training system or fault card set, run a commissioning sequence, identify one fault, and deliver a final technical handoff.



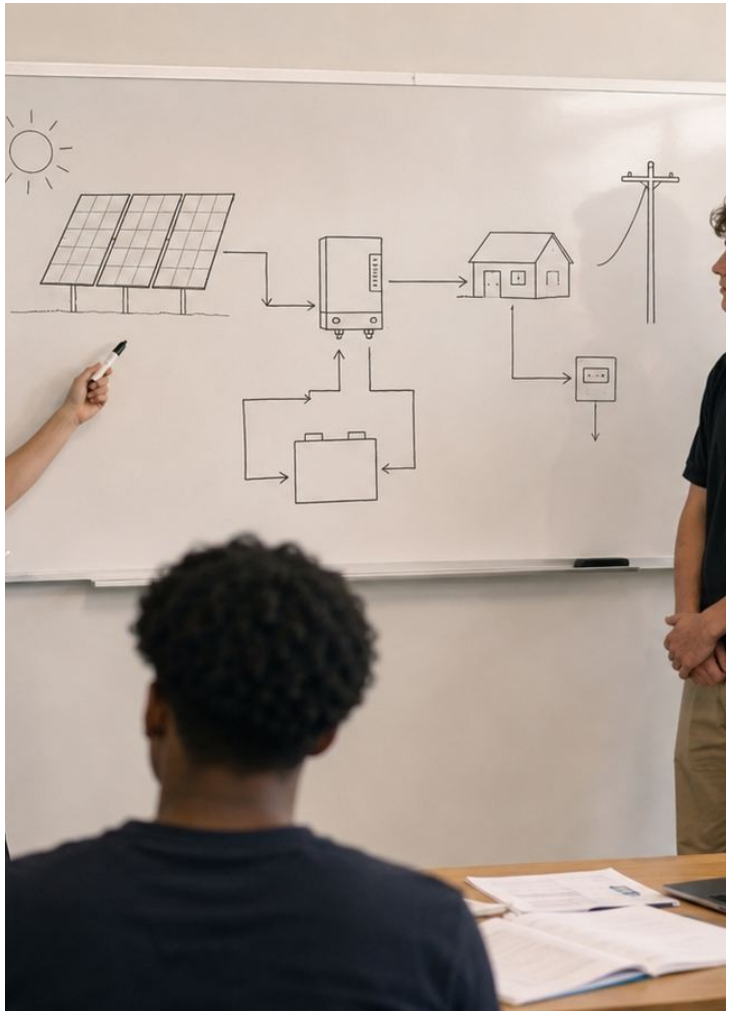
This is the work block. Keep it practical and evidence-based.

Test, document, and hand off what the system actually does.

Scenario options

Assign one, or let teams choose if the room can handle democracy today.

- Training kit has a hidden open fuse and incomplete label.
- Paper system has reversed polarity at one component card and missing OCPD label.
- Mock client handoff has incomplete load table and no record of final voltage checks.

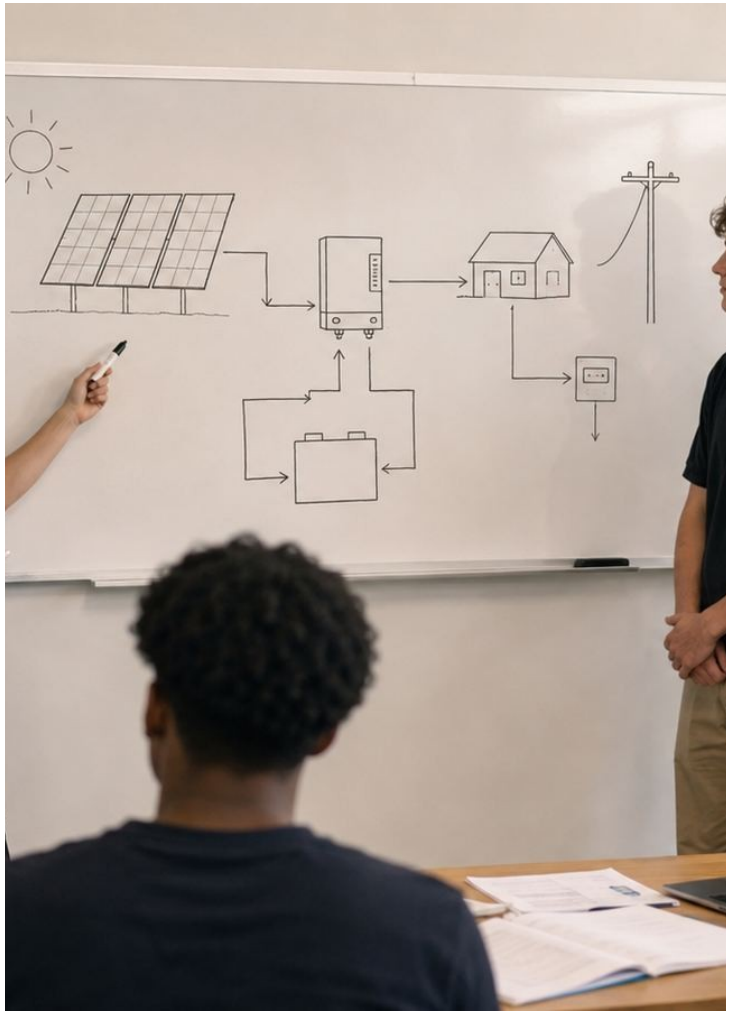


Test, document, and hand off what the system actually does.

Lab procedure

Run the same loop every day.

- Review the system scope and safety boundary.
- Complete pre-energization inspection using the checklist.
- Run only authorized de-energized checks first.
- Use evidence to identify the fault or missing documentation.
- Record pass/fail results and corrective action recommendation.

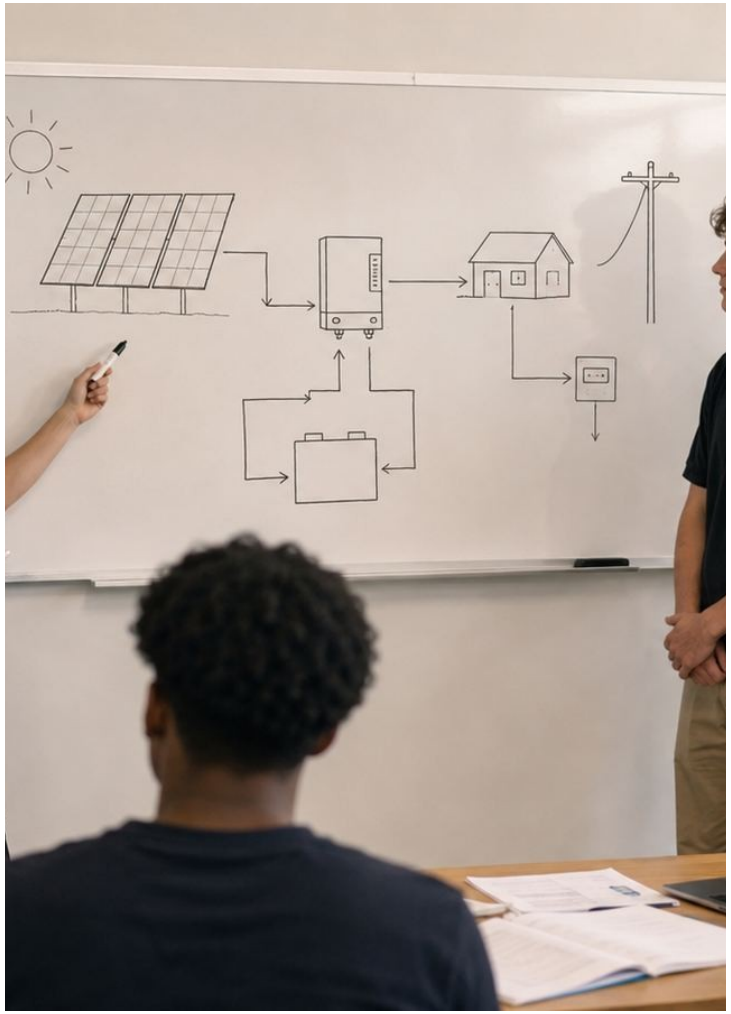


Test, document, and hand off what the system actually does.

Acceptance criteria

A complete answer has five parts.

- Match the need.
- Show the path.
- Name the dominant constraint.
- State the safety boundary.
- Record unknowns instead of guessing.



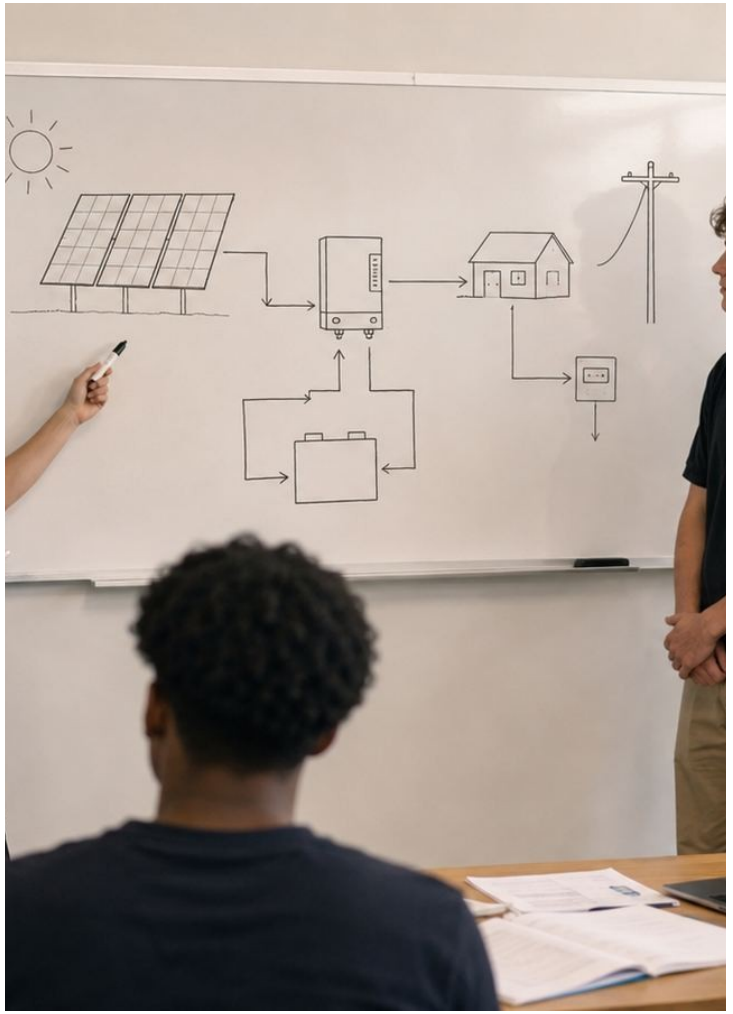
Test, document, and hand off what the system actually does.

Safety boundary

No training objective is improved by pretending energy is harmless.

- Use only instructor-approved equipment and procedures.
- Treat PV and batteries as energy sources.
- Stop when measurements do not make sense.
- Follow manufacturer instructions and program safety procedures.

Stop, isolate, verify, ask. The glamorous life.

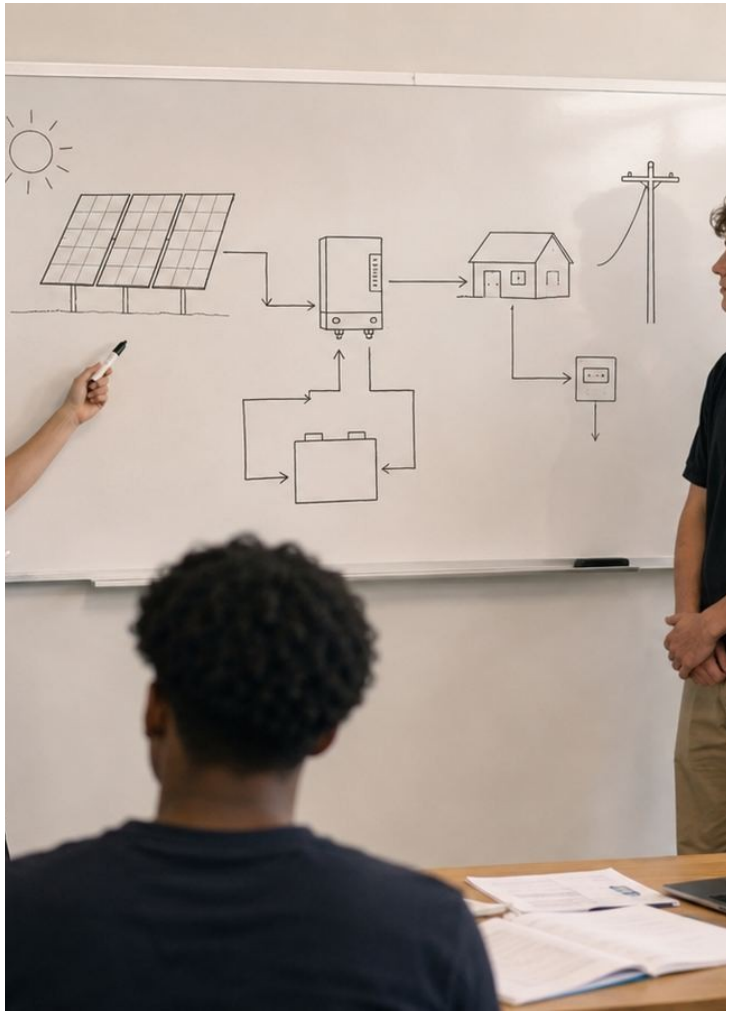


Test, document, and hand off what the system actually does.

Instructor probe questions

Use these when teams get vague.

- What evidence supports that?
- What would make this design fail?
- Which boundary protects the person or equipment?
- What unknown are you carrying forward?
- What would the next crew need to know?

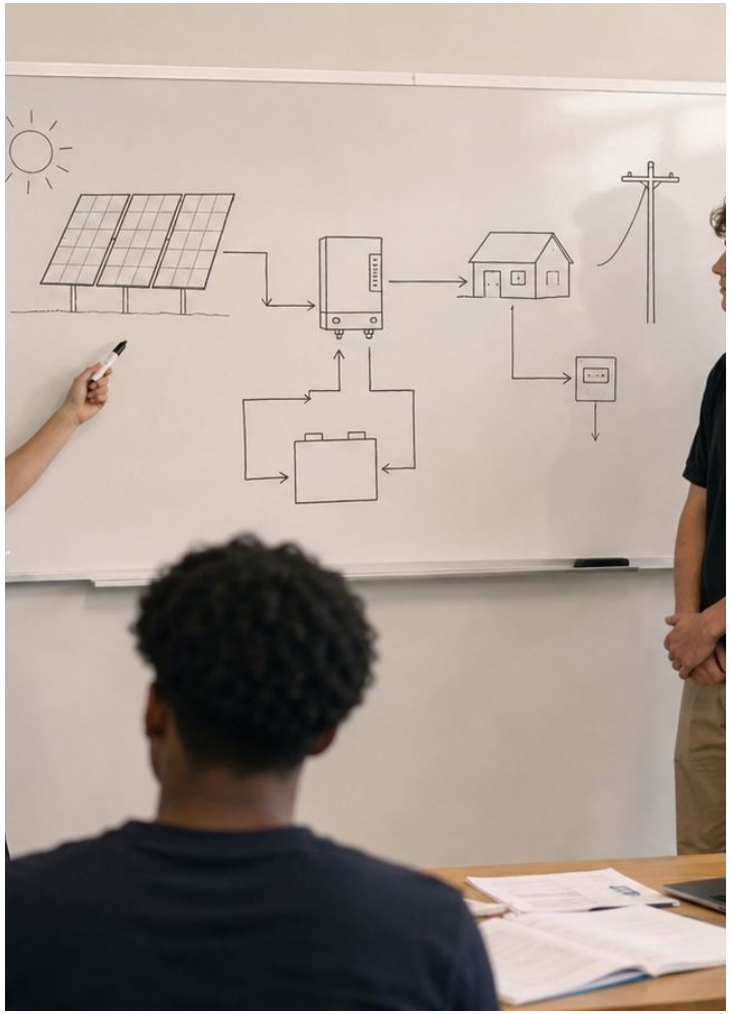


Test, document, and hand off what the system actually does.

Peer verification

Peers check the handoff, not the personality of the speaker. Humanity struggles here.

- Can I identify the job?
- Can I follow the path?
- Can I name the constraint?
- Can I see the safety boundary?
- Can I find the unknowns?

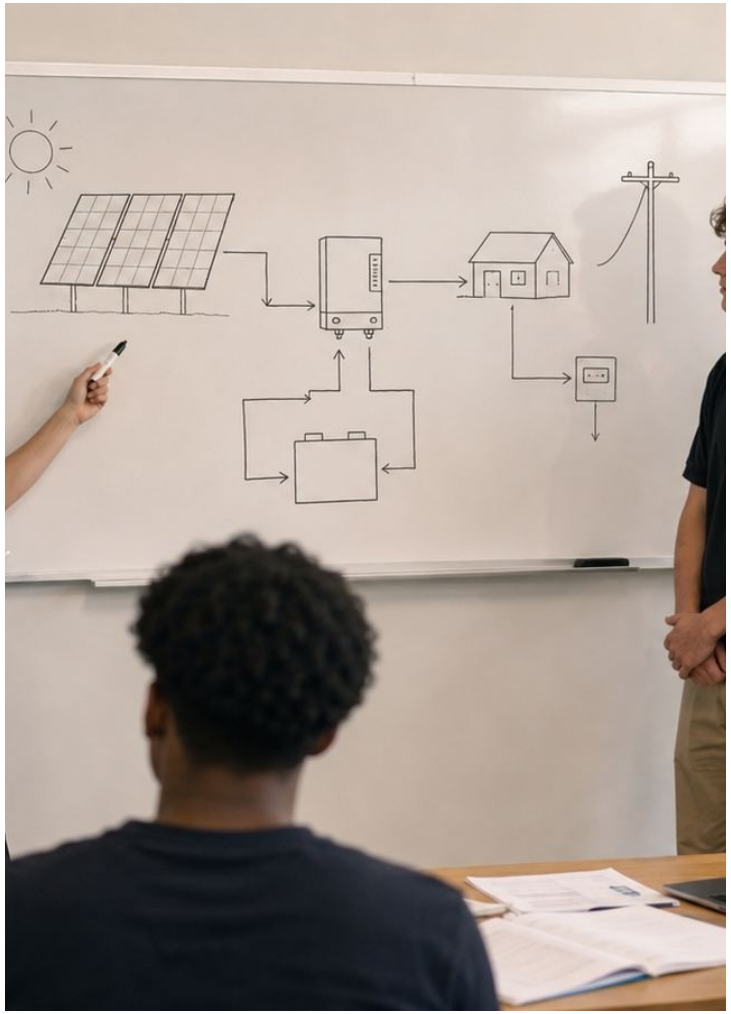


Test, document, and hand off what the system actually does.

Common failure modes

Correct these early.

- Component list with no job.
- Guess presented as fact.
- Missing safety boundary.
- No units or runtime.
- Handoff too vague to use.

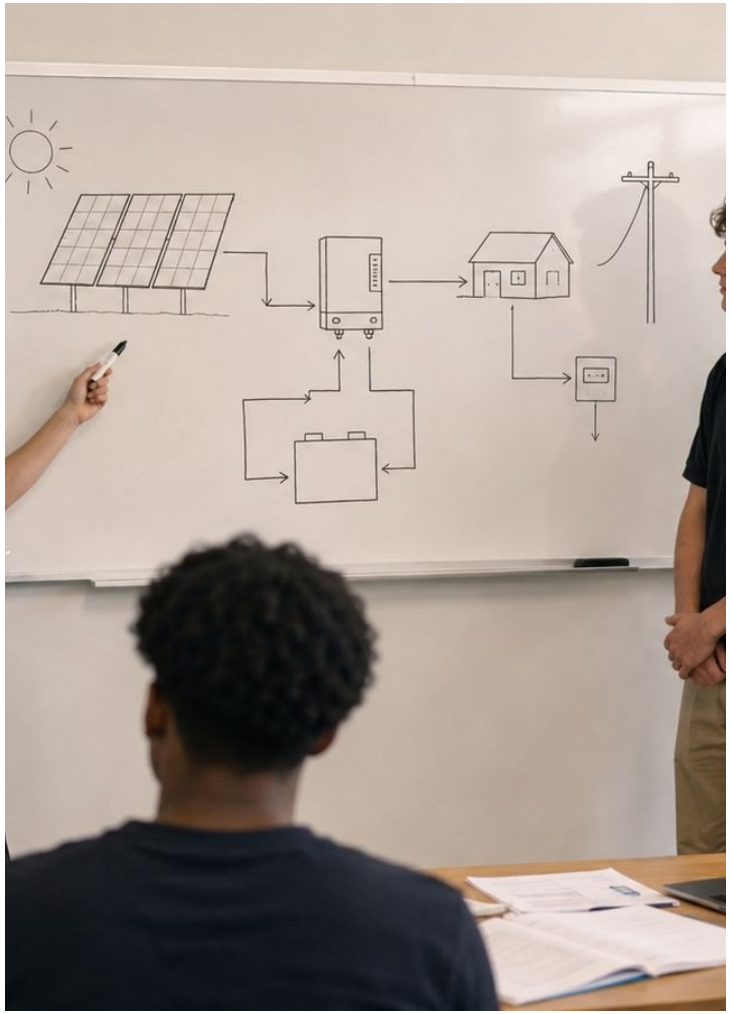


Test, document, and hand off what the system actually does.

Evidence language

Require technical sentences.

- Observed: what you saw.
- Measured: value with unit or clear method.
- Assumed: guess being carried forward.
- Verified: checked against source, procedure, or measurement.

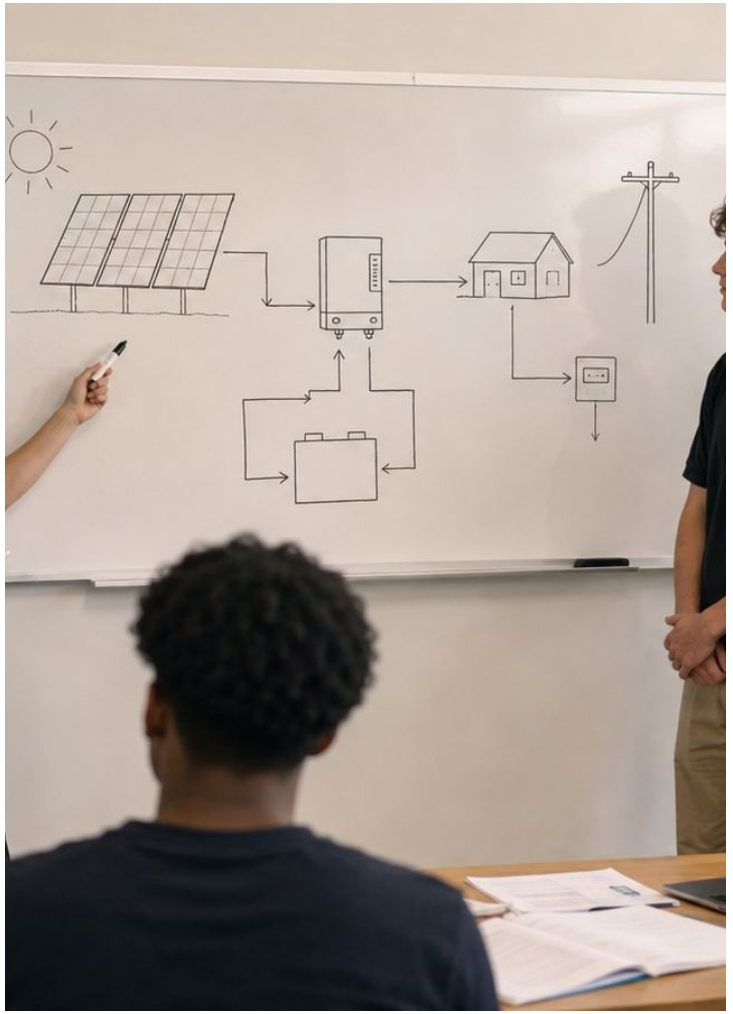


Test, document, and hand off what the system actually does.

Handoff structure

Make the next person safer.

- Scope: what job is being solved.
- Decision: what was chosen or rejected.
- Evidence: why.
- Boundary: what must be protected.
- Unknowns: what still needs verification.

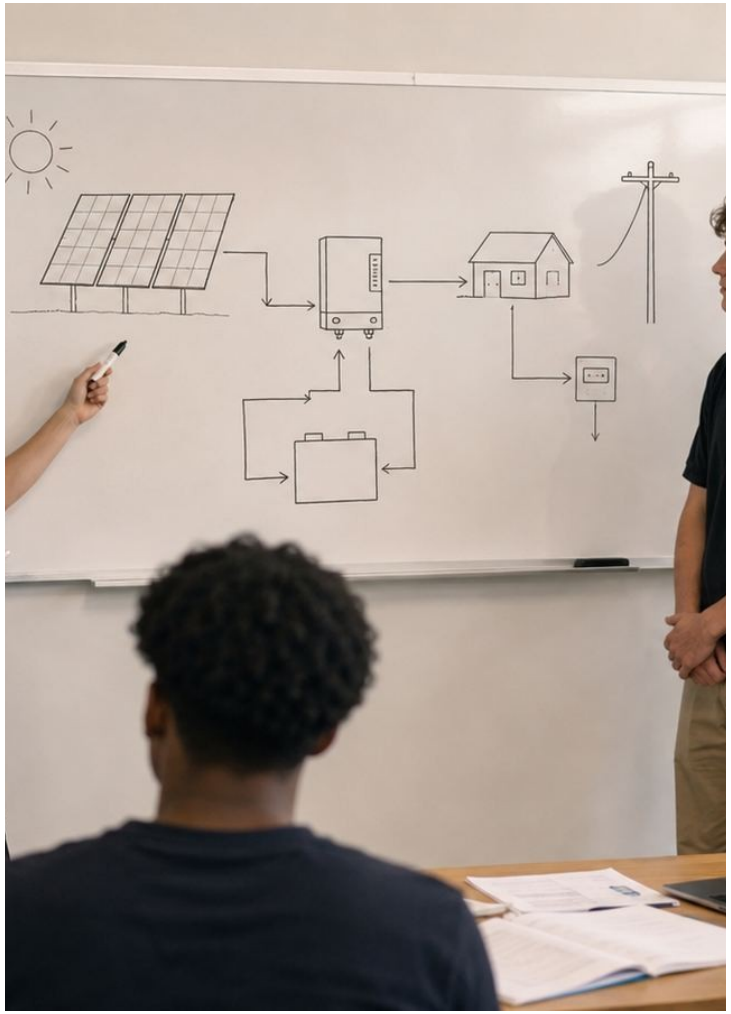


Test, document, and hand off what the system actually does.

Exit ticket

Individual close-out

- What is one thing you now refuse to accept as “good enough” without evidence?

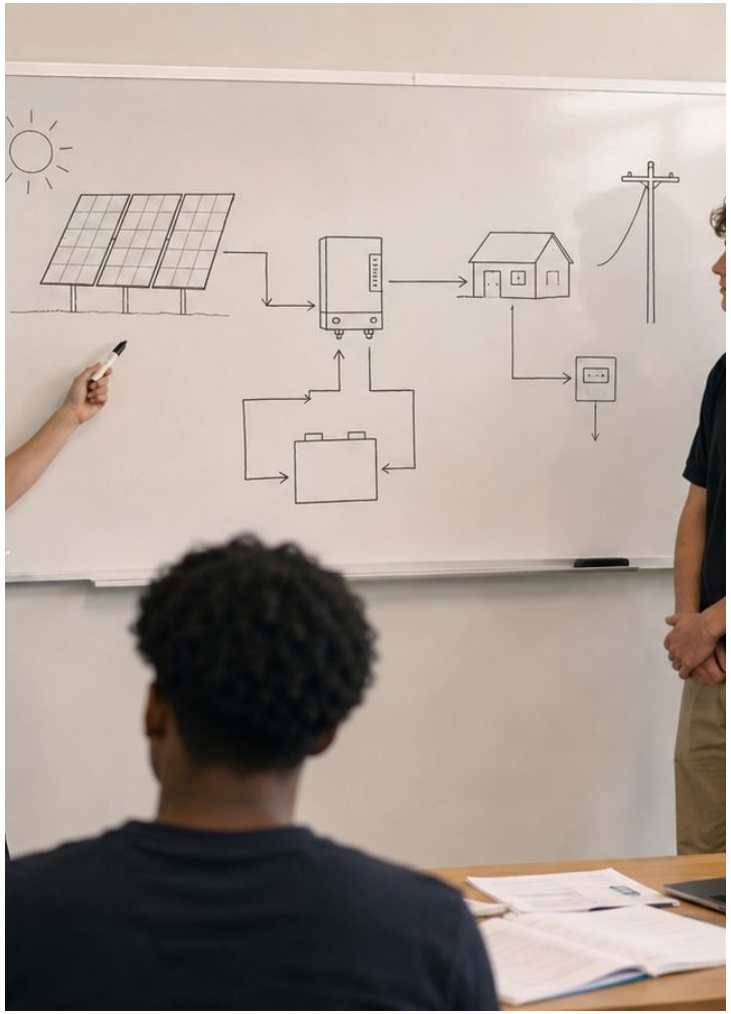


Collect before reset so the day leaves a paper trail. Weirdly useful.

Day-to-day continuity

This is one system, not seven disconnected worksheets.

- PV102 is complete. Carry the habit forward: define the job, verify the path, protect the boundary, document the truth.
- Carry forward site notes, load tables, diagrams, and unknowns.
- Do not erase uncertainty. Resolve it.



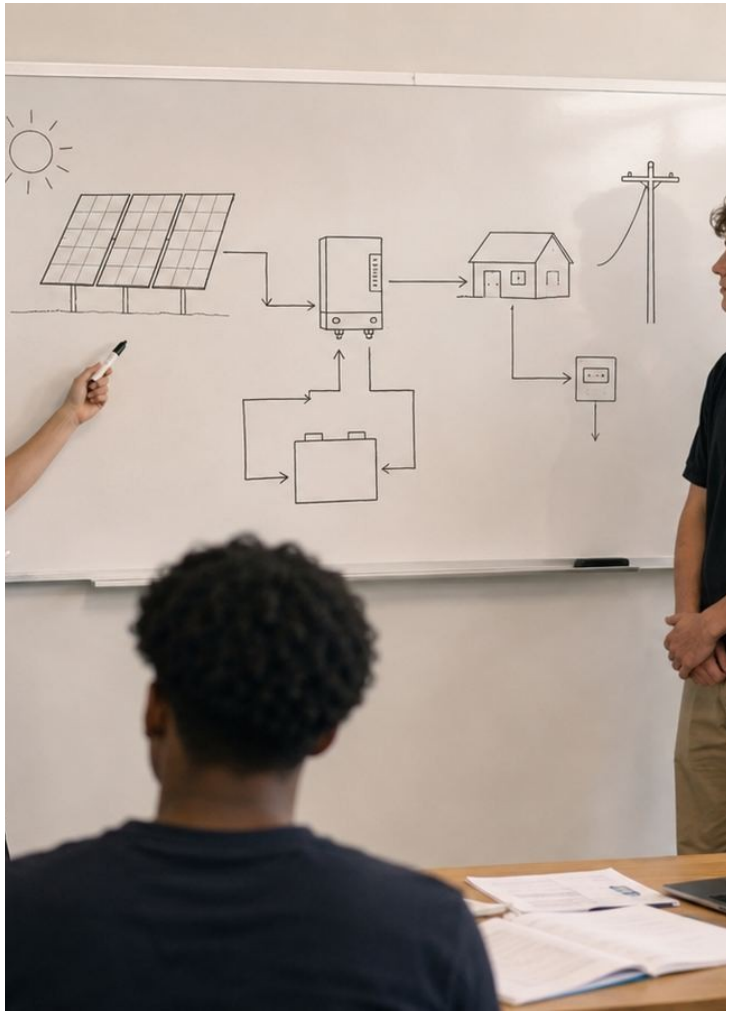
Test, document, and hand off what the system actually does.

Final reminder

What professionals do

- Define the job.
- Verify the path.
- Protect the boundary.
- Document the truth.

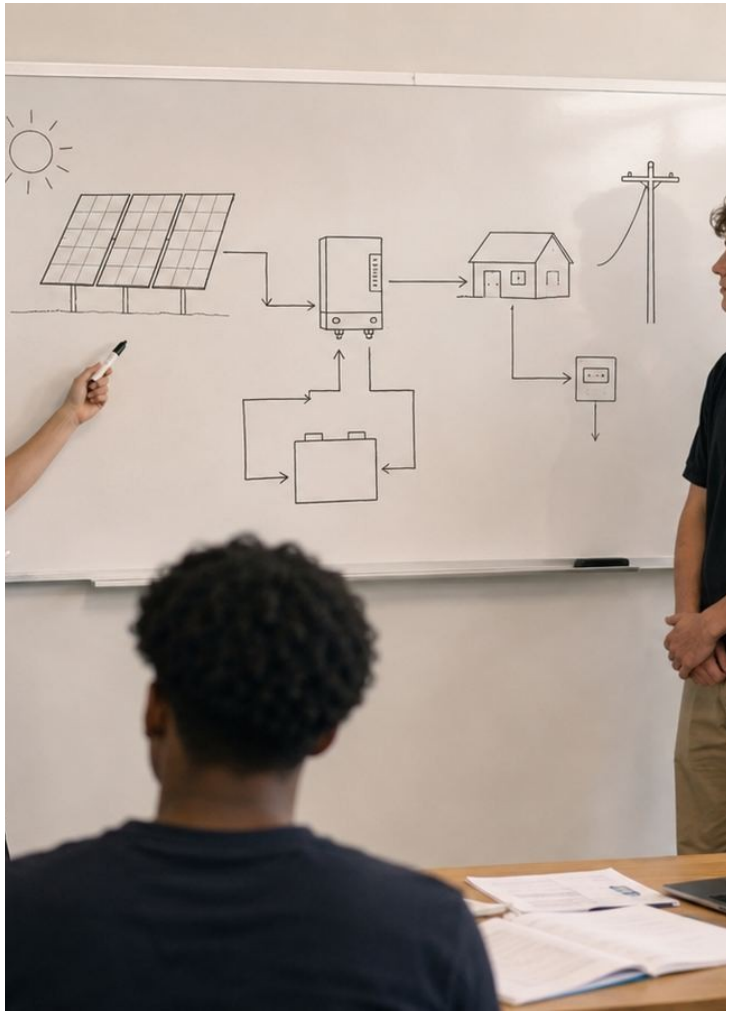
Test, document, and hand off what the system actually does.



Test, document, and hand off what the system actually does.

Close

- Test, document, and hand off what the system actually does.



Reset tools. Collect packets. Archive the handoffs.

Test, document, and hand off what the system actually does.